



```
In [1]: import pandas as pd
```

```
In [2]: df = pd.read_csv("laptop.csv")
```

```
In [3]: df.head()
```

```
Out[3]:
```

	Unnamed: 0	Model	Price	Rating	Generation	Core	Ram	SSD
0	0	HP Victus 15-fb0157AX Gaming Laptop (AMD Ryzen...	₹50,399	70.0	5th Gen AMD Ryzen 5 5600H	Hexa Core, 12 Threads	8 GB DDR4 RAM	512 GB SSD
1	1	Tecno Megabook T1 Laptop (11th Gen Core i3/ 8G...	₹23,990	56.0	11th Gen Intel Core i3 1115G4	Dual Core, 4 Threads	8 GB LPDDR4 RAM	512 GB SSD
2	2	Lenovo V15 G4 82YU00W7IN Laptop (AMD Ryzen 3 ...	₹26,989	55.0	7th Gen AMD Ryzen 3 7320U	Quad Core, 8 Threads	8 GB LPDDR5 RAM	512 GB SSD
3	3	Samsung Galaxy Book2 Pro 13 Laptop (12th Gen C...	₹69,990	60.0	12th Gen Intel Core i5 1240P	12 Cores (4P + 8E), 16 Threads	16 GB LPDDR5 RAM	512 GB SSD
4	4	Xiaomi Redmi G Pro 2024 Gaming Laptop (14th Ge...	₹1,02,990	78.0	14th Gen Intel Core i9 14900HX	24 Cores (8P + 16E), 32 Threads	16 GB DDR5 RAM	1 TB SSD

```
In [4]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 920 entries, 0 to 919
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Unnamed: 0      920 non-null   int64
1   Model           920 non-null   object
2   Price           920 non-null   object
3   Rating          768 non-null   float64
4   Generation      920 non-null   object
5   Core            920 non-null   object
6   Ram             920 non-null   object
7   SSD             920 non-null   object
8   Display         919 non-null   object
9   Graphics        920 non-null   object
10  OS              919 non-null   object
11  Warranty        915 non-null   object
dtypes: float64(1), int64(1), object(10)
memory usage: 86.4+ KB

```

```
In [5]: df.describe()
```

```
Out[5]:
```

	Unnamed: 0	Rating
count	920.000000	768.000000
mean	459.500000	61.756510
std	265.725422	7.114217
min	0.000000	50.000000
25%	229.750000	56.000000
50%	459.500000	61.500000
75%	689.250000	66.000000
max	919.000000	79.000000

```
In [6]: df.isnull().sum()
```

Out[6]:

	0
Unnamed: 0	0
Model	0
Price	0
Rating	152
Generation	0
Core	0
Ram	0
SSD	0
Display	1
Graphics	0
OS	1
Warranty	5

dtype: int64

```
In [7]: df.duplicated().sum()
```

Out[7]: np.int64(0)

```
In [8]: df = df.drop_duplicates()
```

```
In [9]: df = df.drop("Unnamed: 0", axis=1)
```

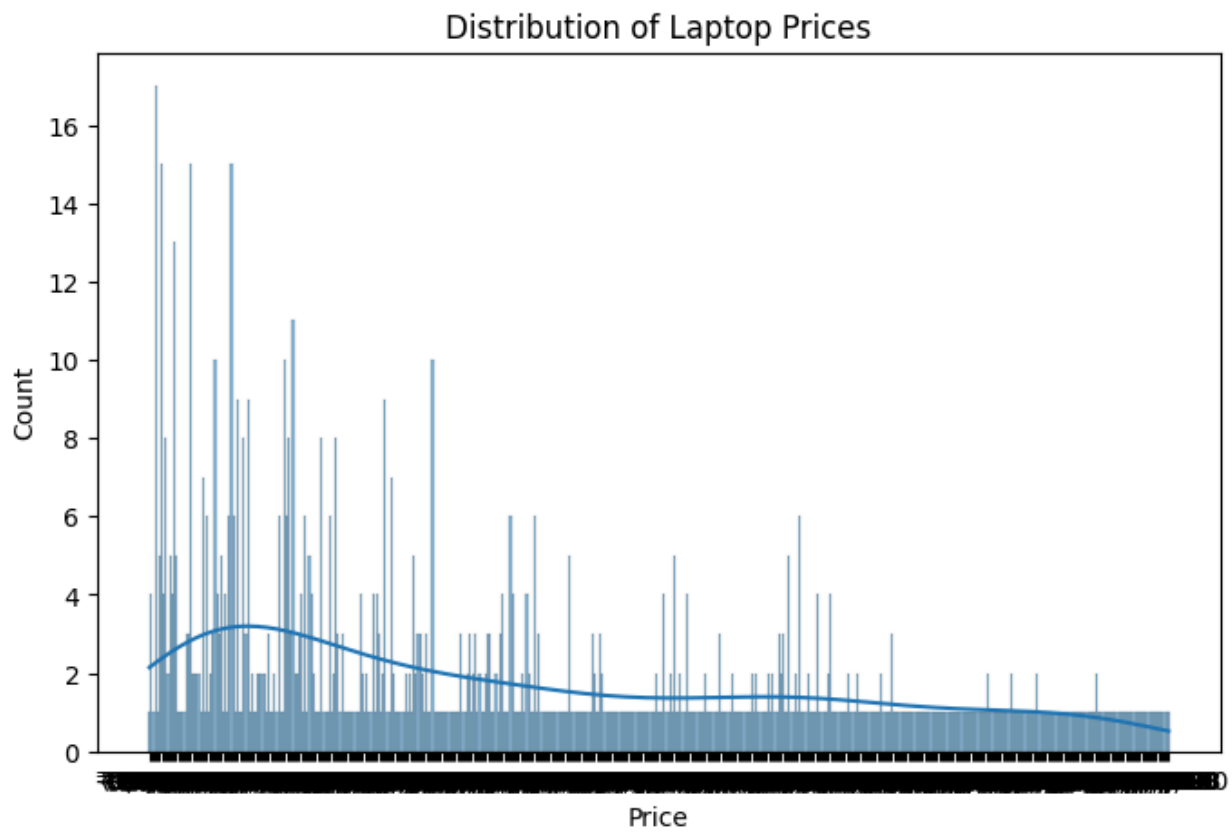
```
In [10]: df.head()
```

Out[10]:

	Model	Price	Rating	Generation	Core	Ram	SSD	Display	Gr
0	HP Victus 15-fb0157AX Gaming Laptop (AMD Ryzen...	₹50,399	70.0	5th Gen AMD Ryzen 5 5600H	Hexa Core, 12 Threads	8 GB DDR4 RAM	512 GB SSD	15.6 inches, 1920 x 1080 pixels, Touch Screen	4 C
1	Tecno Megabook T1 Laptop (11th Gen Core i3/ 8G...	₹23,990	56.0	11th Gen Intel Core i3 1115G4	Dual Core, 4 Threads	8 GB LPDDR4 RAM	512 GB SSD	15.6 inches, 1920 x 1080 pixels	In C
2	Lenovo V15 G4 82YU00W7IN Laptop (AMD Ryzen 3 ...	₹26,989	55.0	7th Gen AMD Ryzen 3 7320U	Quad Core, 8 Threads	8 GB LPDDR5 RAM	512 GB SSD	15.6 inches, 1920 x 1080 pixels	C
3	Samsung Galaxy Book2 Pro 13 Laptop (12th Gen C...	₹69,990	60.0	12th Gen Intel Core i5 1240P	12 Cores (4P + 8E), 16 Threads	16 GB LPDDR5 RAM	512 GB SSD	13.3 inches, 1080 x 1920 pixels	C
4	Xiaomi Redmi G Pro 2024 Gaming Laptop (14th Ge...	₹1,02,990	78.0	14th Gen Intel Core i9 14900HX	24 Cores (8P + 16E), 32 Threads	16 GB DDR5 RAM	1 TB SSD	16.1 inches, 2560 x 1600 pixels	C R

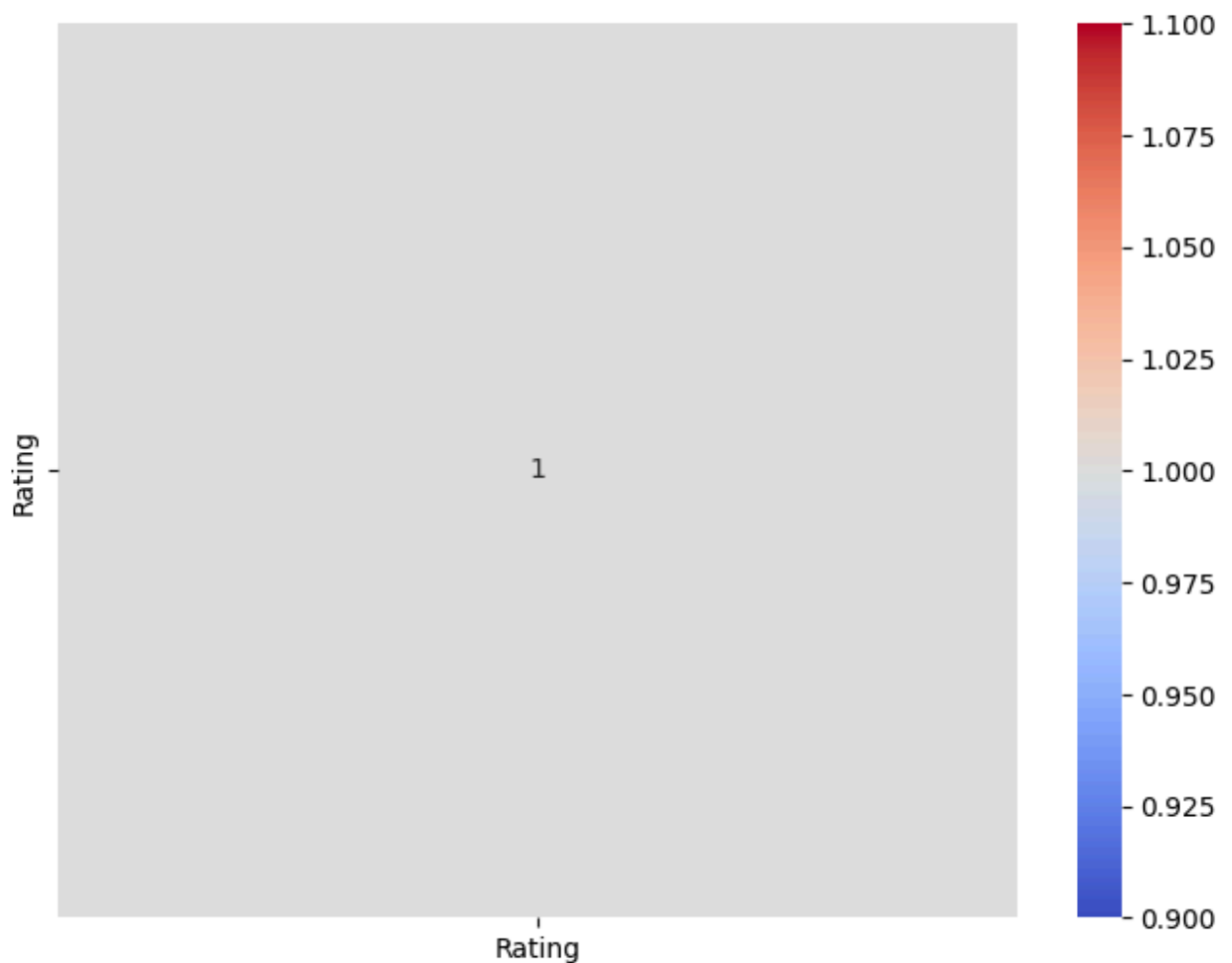
```
In [11]: import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [12]: plt.figure(figsize=(8,5))
sns.histplot(df["Price"], bins=30, kde=True)
plt.title("Distribution of Laptop Prices")
plt.show()
```



```
In [13]: numeric_df = df.select_dtypes(include=np.number)

plt.figure(figsize=(8,6))
sns.heatmap(numeric_df.corr(), annot=True, cmap="coolwarm")
plt.show()
```



```
In [14]: from sklearn.preprocessing import LabelEncoder  
  
le = LabelEncoder()  
  
for col in df.select_dtypes(include="object").columns:  
    df[col] = le.fit_transform(df[col])
```

```
In [15]: X = df.drop("Price", axis=1)  
y = df["Price"]
```

```
In [16]: from sklearn.model_selection import train_test_split  
  
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.2,  
    random_state=42  
)
```

```
In [17]: from sklearn.ensemble import RandomForestRegressor  
  
model = RandomForestRegressor(random_state=42)
```

```
model.fit(X_train, y_train)
```

Out[17]:

▼ RandomForestRegressor ⓘ ?
RandomForestRegressor(random_state=42)

In [18]: `y_pred = model.predict(X_test)`

In [19]: `from sklearn.metrics import mean_absolute_error
from sklearn.metrics import mean_squared_error
from sklearn.metrics import r2_score

mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = mse ** 0.5
r2 = r2_score(y_test, y_pred)

print("MAE:", mae)
print("MSE:", mse)
print("RMSE:", rmse)
print("R2 Score:", r2)`

MAE: 62.348260869565216
MSE: 10220.12485326087
RMSE: 101.09463315755625
R2 Score: 0.4370464457537948